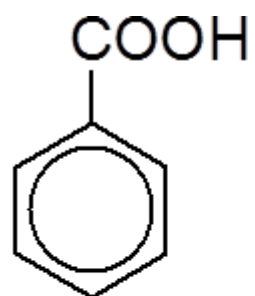
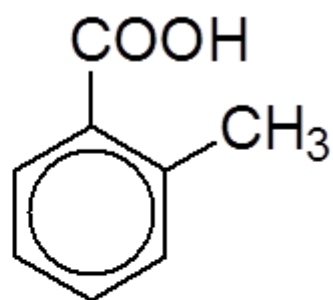


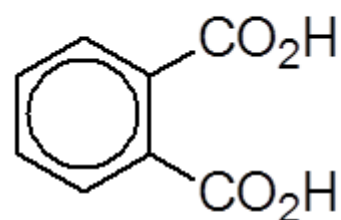
Aromatic Carboxylic acids



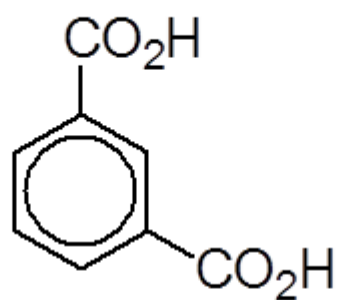
benzoic acid



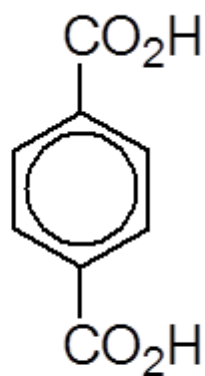
o-toluic acid



phthalic acid



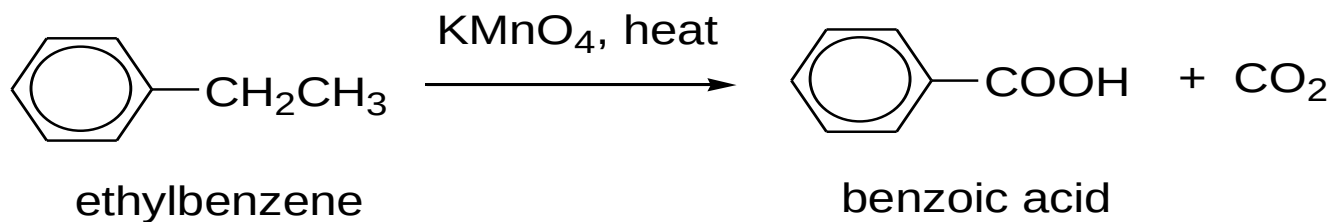
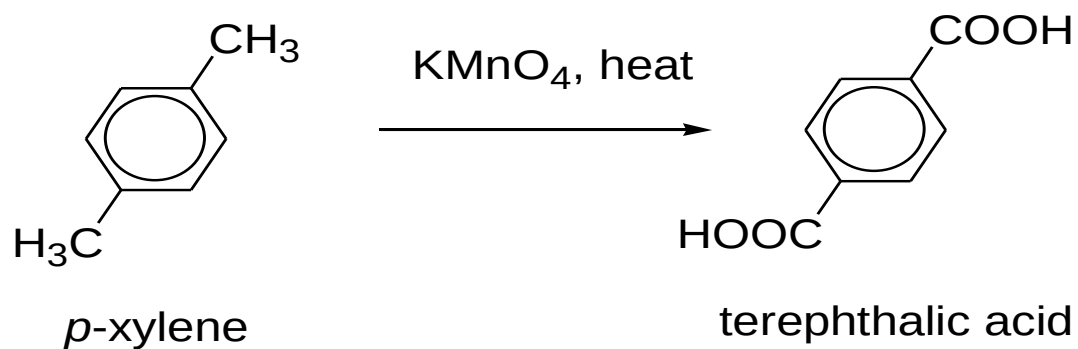
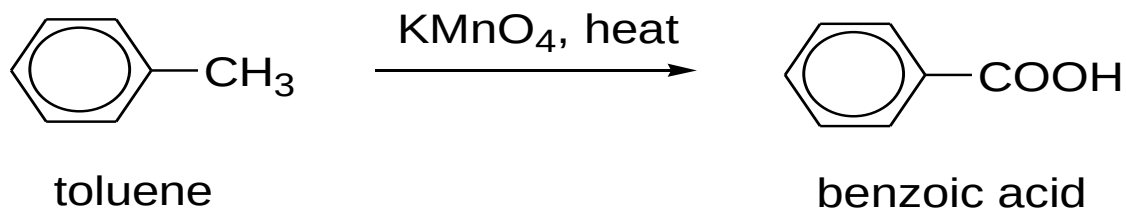
isophthalic acid



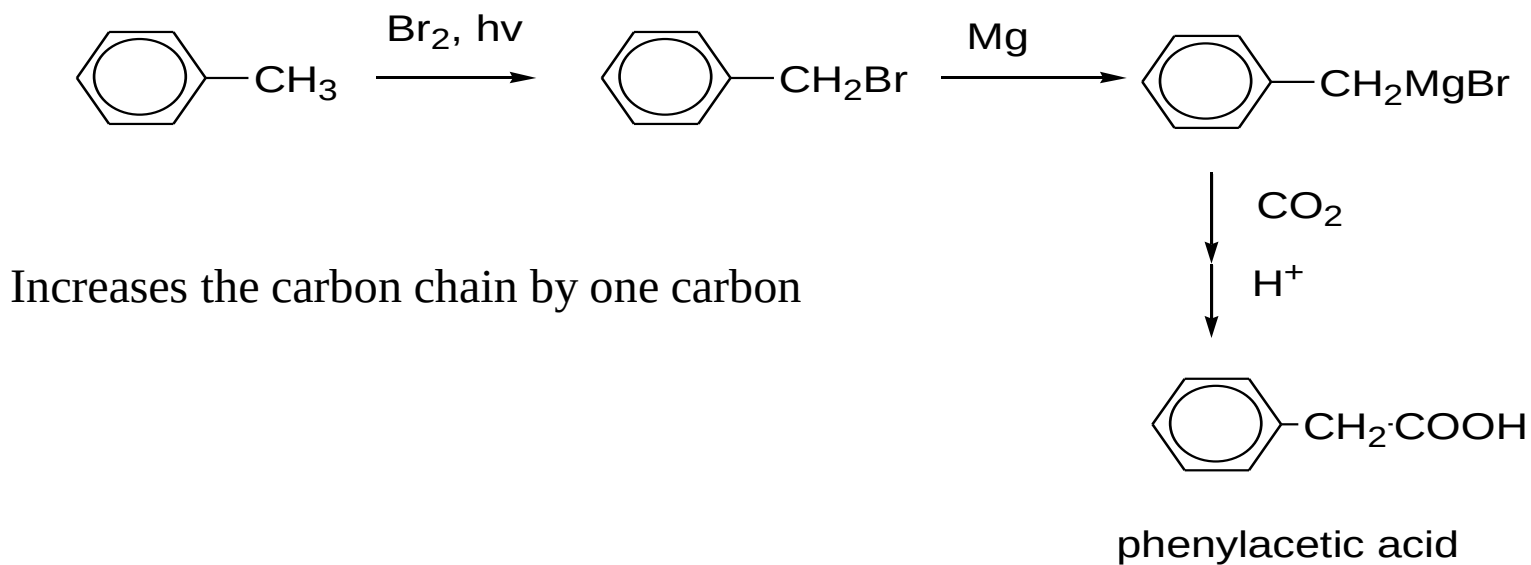
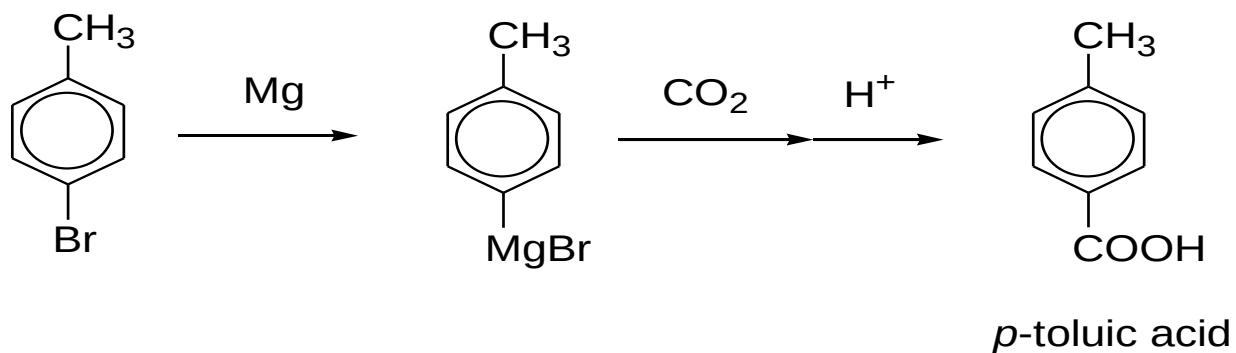
terephthalic acid

Synthesis of Aromatic Carboxylic acids

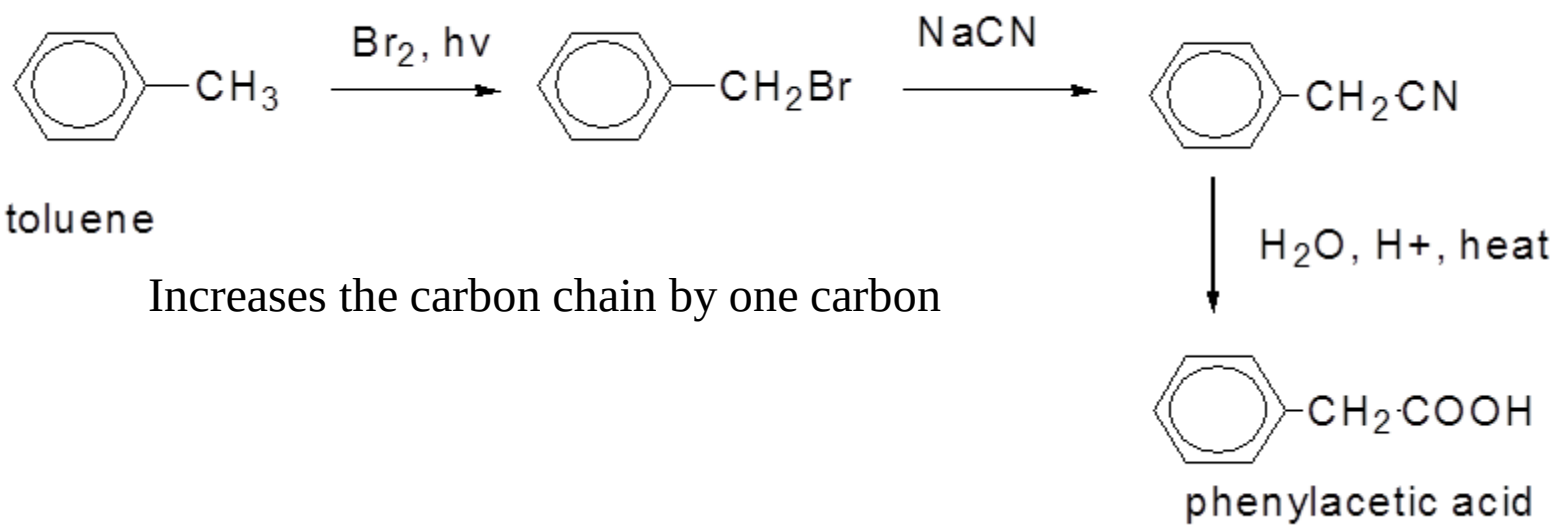
1. Oxidation of arenes:



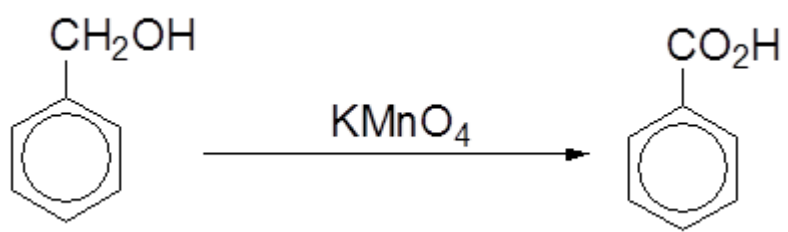
2. Carbonation of Grignard reagent:



3. Hydrolysis of a nitrile



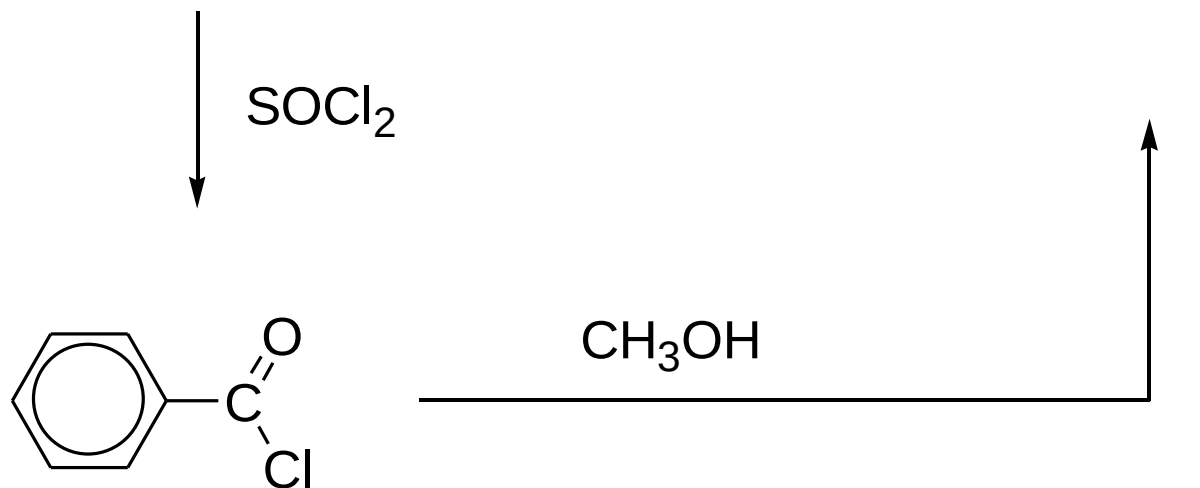
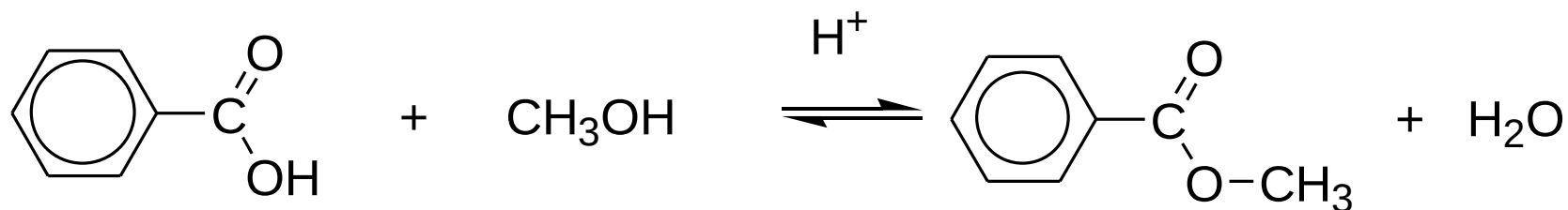
4. Oxidation of alcohols



Reactions of Aromatic carboxylic acids

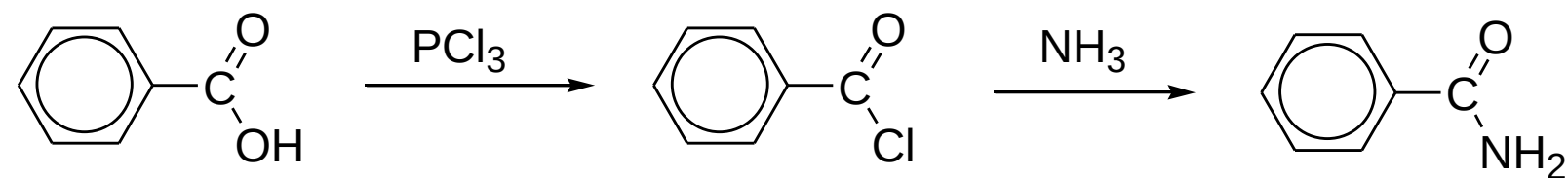
1. Esterification

“direct” esterification - reversible

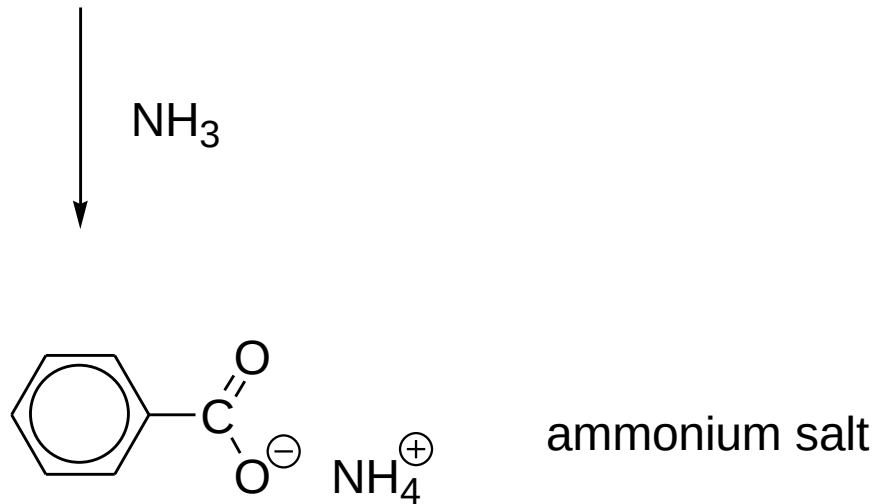


“indirect” esterification - irreversible

2. Formation of amides



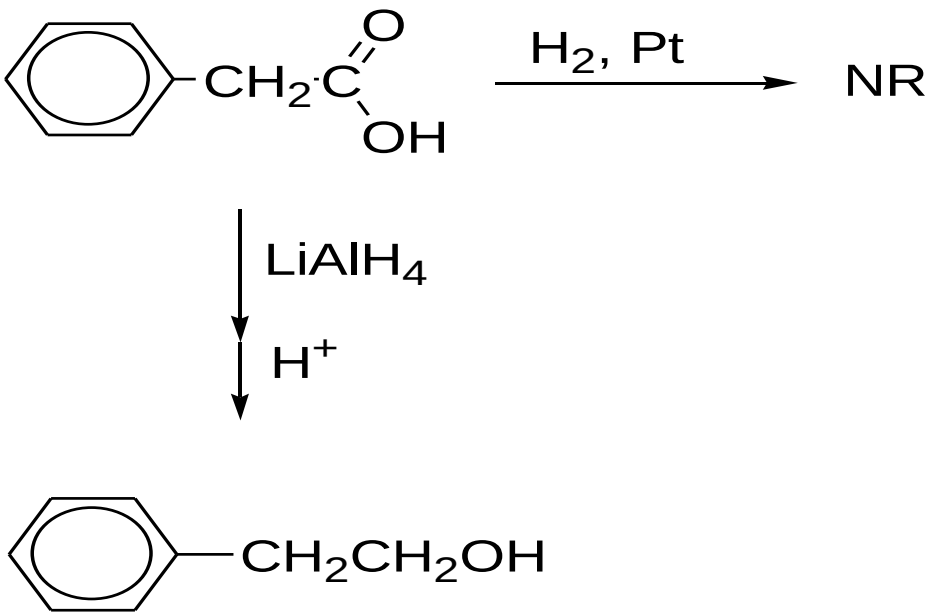
amide



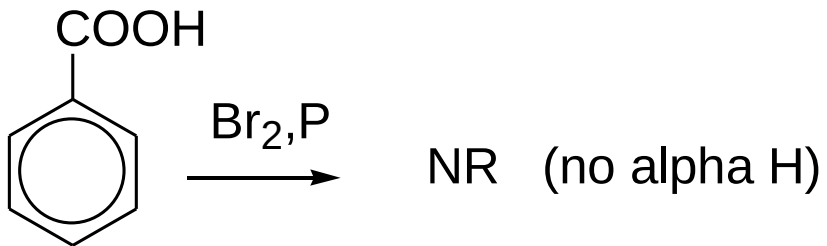
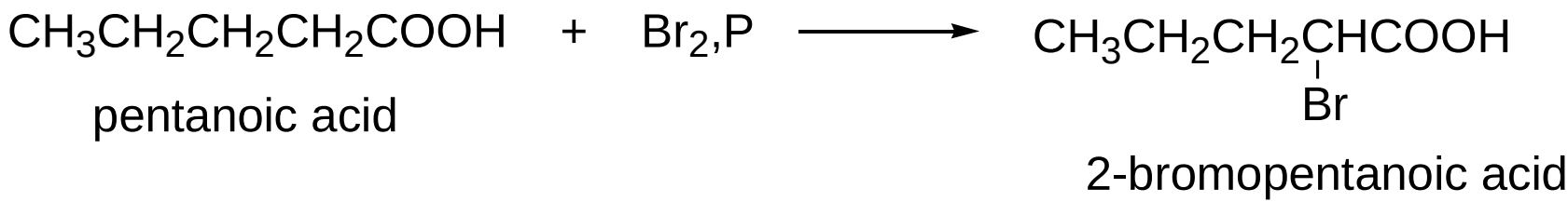
ammonium salt

3. Reduction

Carboxylic acids resist catalytic reduction under normal conditions.



4. Alpha-halogenation: (Hell-Volhard-Zelinsky reaction)



5. EAS: (-COOH is deactivating and *meta*- directing)

